

**Impact of Tax Exemptions on
Dependent Children: A Time Series
Analysis of US Households
(1913 - 1984)**

**ECO348-FAC413-DOM637 - Introduction to
Mathematical Finance and Financial
Engineering**

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1. Introduction:

This study examines how fiscal policy shaped fertility decisions in the United States from 1913 to 1984. Specifically, it investigates whether real tax exemptions for dependents influenced fertility decisions. The dependent variable is the number of children aged 0-3 per 1,000 women (*chld*), while the key explanatory variable is the real dollar value of tax exemptions (*exem*).

Since fertility decisions are unlikely to adjust instantaneously to changes in financial incentives, the analysis considers both short-run and longer-term responses. The study also accounts for major historical influences, including World War II (1941-1945) and the introduction of a government educational program for women in 1963, to isolate the effect of tax policy more precisely.

2. Methodology:

2.1 Data:

The analysis uses annual time-series data for the United States from 1913 to 1984, drawn from the FERTIL3 dataset contained in the *FemaleEmploy.xlsx* file. The dataset consists of 72 observations, reduced to 67 after constructing lagged variables.

Key variables include *chld* (dependent children aged 0–3 per 1,000 women, used for fertility), *exem* (real dollar value of tax exemptions), *wartm* (dummy for World War II, 1941–1945), *eduprog* (dummy for the women’s educational program introduced in 1963), and *t* (time trend, with t^2 and t^3 included to capture nonlinear trends).

All variables are converted to numeric format, sorted chronologically, and observations lost due to lag construction are dropped prior to estimation.

2.2 Variable Construction:

To capture delayed responses to tax policy, four lags of *exem* (*exem*₁–*exem*₄) are generated. First differences are computed as:

$$\begin{aligned}\Delta chld_t &= chld_t - chld_{t-1} \\ \Delta exem_t &= exem_t - exem_{t-1}\end{aligned}$$

Lagged values of these differenced variables (up to four periods) are also constructed. These transformations enable both dynamic level models and short-run difference-based specifications.

2.3 Stationarity Testing:

The Augmented Dickey-Fuller (ADF) test is applied to both levels and first differences of *chld* and *exem* to determine their order of integration. Levels are tested with a constant and trend, while first differences are tested with a constant only. The results guide the choice between levels-based and differenced specifications.

2.4 Model Design and Specification:

Trend Models:

Linear, quadratic, and cubic time trends are estimated to capture deterministic patterns in fertility. The specification that provides the best statistical fit and significance is retained in subsequent models.

Static Model:

The baseline model is:

$$chld_t = \beta_0 + \beta_1 exem_t + \beta_2 wartm_t + \beta_3 eduprogt + f(t) + u_t$$

where $f(t)$ represents polynomial time trends.

Finite Distributed Lag (FDL) Model:

To capture dynamic effects, the model is extended to include current and lagged exemptions:

$$chld_t = \beta_0 + \beta_1 exem_t + \beta_2 exem_{t-1} + \beta_3 exem_{t-2} + \beta_4 exem_{t-3} + \beta_5 exem_{t-4} + \gamma X_t + f(t) + u_t$$

The Long-Run Propensity (LRP) is defined as:

$$LRP = \beta_1 + \beta_2 + \beta_3 + \beta_4 + \beta_5$$

which measures the cumulative effect of a sustained change in tax exemptions. The standard error of the LRP is computed using the variance–covariance matrix of the estimated lag coefficients, by summing all corresponding variance and covariance terms (delta method).

Changes-Based Model:

As a robustness check, the model is estimated in first differences:

$$\Delta chld_t = \alpha_0 + \sum_{i=0}^4 \alpha_i \Delta exem_{t-i} + \delta X_t + \varepsilon_t$$

This specification focuses on short-run responses and is appropriate if the variables are difference-stationary.

2.5 Diagnostics and Model Evaluation:

Before estimation, stationarity and appropriate functional form are assessed to ensure that key assumptions of the Classical Linear Model are not violated.

CLM assumptions are considered through model specification and assessed post-estimation. The Durbin-Watson and Breusch-Godfrey tests detect autocorrelation, while the Breusch-Pagan test assesses heteroskedasticity.

Model performance is evaluated using R^2 , Adjusted R^2 , AIC, and BIC. A joint F-test is also conducted to assess the significance of lagged exemption terms in the FDL model.

3. Results and Discussions:

3.1 Descriptive Analysis and Time Trends:

Table 1: Descriptive Statistics:

Statistic	chld	exem
count	72.00	72.00
mean	95.63	100.40
std	19.80	65.88
min	65.40	0.00
median	94.05	121.22
max	126.60	243.83

The average number of dependent children (chld) is 95.63 per 1,000 women, ranging from 65.4 to 126.6, indicating substantial variation in fertility over time. Real tax exemptions (exem) average \$100.40, with a wide range from \$0 to \$243.83, reflecting the expansion of fiscal support.

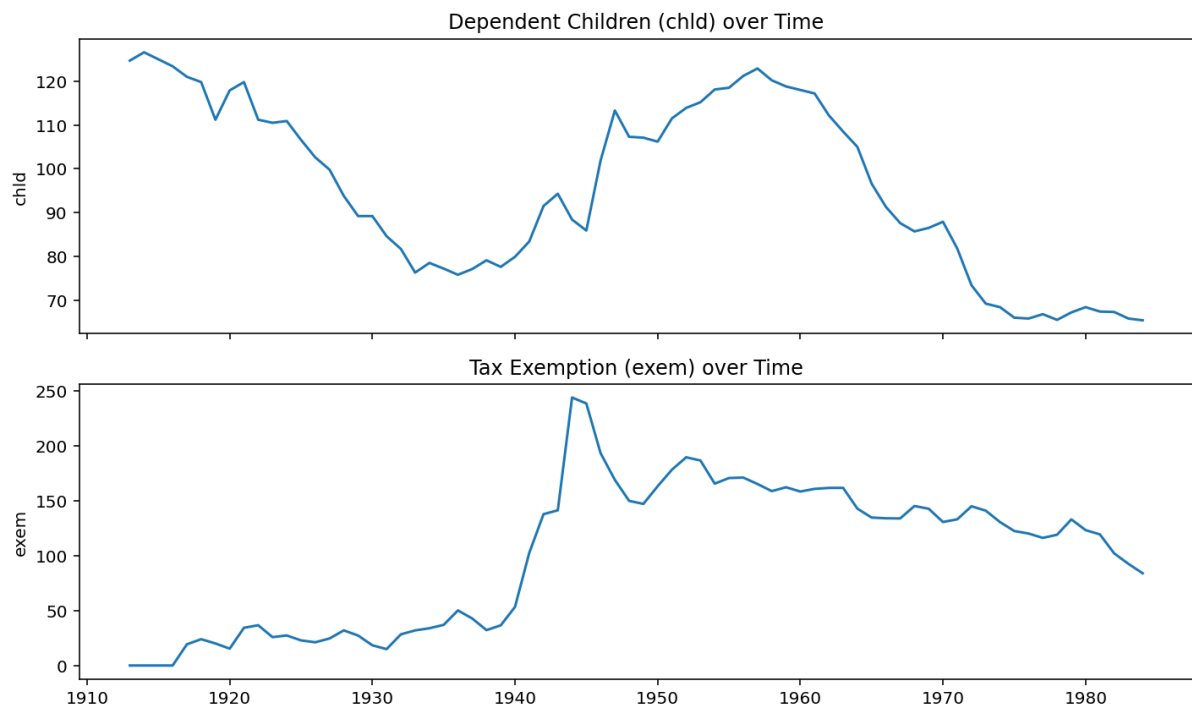


Figure 1: Dependent Children vs Time

Figure 2: Tax Exemption vs Time

The time-series plots show that *chld* follows a non-linear pattern, declining in the early period, rising around the mid-century, and falling thereafter. In contrast, *exem* remains low initially, increases sharply from the 1940s, and stays elevated in later years. The lack of close contemporaneous movement between the two series suggests that the effect of tax exemptions on fertility may operate with lags.

3.2 Stationarity and Trend Analysis:

Table 2: ADF Test Results

Variable	Specification	Test Statistic	p-value	Conclusion
chld	Level (with trend)	-2.839	0.183	Non-stationary
exem	Level (with trend)	-1.471	0.839	Non-stationary
Δ chld	First difference	-2.671	0.079	Borderline (stationary at 10% level)
Δ exem	First difference	-6.486	0.000	Stationary

ADF tests indicate that both chld ($p = 0.183$) and exem ($p = 0.839$) are non-stationary in levels. In first differences, Δ exem is stationary ($p = 0.000$), while Δ chld is borderline stationary (significant at the 10% level, $p = 0.079$), indicating persistence in fertility. Accordingly, both levels and differenced specifications are considered.

Table 3: Trend Model Comparison

Model	R^2	Adj. R^2	AIC	BIC	Conclusion
Linear	0.195	0.182	572.2	576.6	Weak fit
Quadratic	0.304	0.282	564.4	571.0	Better fit
Cubic	0.614	0.595	526.9	535.7	Best fit

Trend regressions show that the cubic specification provides the best fit ($R^2 = 0.614$), outperforming linear (0.195) and quadratic (0.304) models. All cubic terms are statistically significant, confirming strong non-linear time effects.

Overall, exem is classified as an integrated process of order one ($I(1)$) and is difference-stationary, while chld is better characterized as trend-stationary (TS), with non-stationarity driven by deterministic time trends rather than stochastic shocks.

3.3 Static Model Results:

Table 4: Static Model Results

Variable	Coefficient	p-value
exem	0.142	0.001
wartm	-15.396	0.004
eduprog	-29.330	0.000

The static model explains a large share of variation in fertility ($R^2 = 0.832$). Tax exemptions have a positive and significant effect (0.1418, $p = 0.001$), implying that a \$12 increase raises fertility by about 1.70 children per 1,000 women, indicating a modest short-run impact.

Both wartm (-15.40 , $p = 0.004$) and eduprog (-29.33 , $p < 0.001$) have significant negative effects. This indicates that fertility was significantly lower during the World War II period, and following the introduction of the education program.

Cubic trend terms (t , t^2 , t^3) and the constant are included in the regression but not reported in the table; all are statistically significant ($p < 0.001$).

Table 5: Static Model Diagnostics

Test	Value
R^2	0.832
Durbin-Watson	0.583
Breusch-Pagan p-value	0.006
Breusch-Godfrey p-value	0.000

However, diagnostics reveal strong autocorrelation ($DW = 0.583$; $BG\ p \approx 0.000$) and heteroskedasticity ($BP\ p = 0.006$), limiting the reliability of inference.

3.4 Finite Distributed Lag (FDL) Model Results:

Table 6: FDL Model Results

Variable	Coefficient	p-value
exem	-0.0202	0.787
exem ₁	-0.0372	0.656
exem ₂	0.1408	0.122
exem ₃	0.0079	0.924
exem ₄	0.0945	0.122
eduprog	-26.915	0.000
warm	4.250	0.529

Including lagged exemptions improves model fit ($R^2 = 0.871$; lower AIC). The joint F-test confirms that lagged terms are significant ($p = 0.004$), indicating dynamic effects.

Lagged exemptions are highly correlated (≈ 0.79 – 0.96), reflecting severe multicollinearity which inflates standard errors and prevents precise estimation of individual effects.

The warm coefficient becomes insignificant, suggesting that lag structure absorbs variation previously captured in the static model.

Table 7: FDL Diagnostics and Joint Test

Test	Value
R^2	0.871
Durbin-Watson	0.558
Breusch-Pagan (p-val)	0.028
Breusch-Godfrey (p-val)	0.000
Joint F-test (lags p-val)	$F = 4.295, p = 0.0042$
Condition Number	1,419,220

Diagnostics still indicate autocorrelation and heteroskedasticity, so econometric issues persist despite improved fit.

3.5 Long-Run and Short-Run Effects:

Table 8: Long-Run Propensity (LRP)

Measure	Value
LRP	0.186
Std. Error	0.042
t-stat	4.475
p-value	0.000

The Long-Run Propensity (LRP) is 0.1859 ($p < 0.001$), indicating that a sustained increase in tax exemptions has a positive cumulative effect on fertility over time.

Table 9: Model Comparison and Changes-Based Model

Metric / Model	Static	FDL	Changes Model
R ²	0.832	0.871	0.336
AIC	477.3	467.4	-
BIC	492.8	491.6	-
Δ exem Coefficient	-	-	-0.095
Δ exem p-value	-	-	0.011
Durbin-Watson	0.583	0.558	1.506
BP p-value	0.006	0.028	0.669
BG p-value	0.000	0.000	0.113

Model comparison favours the FDL specification (higher R², lower AIC), supporting the inclusion of dynamics.

The changes-based model shows a negative short-run effect of exemptions (-0.0953 , $p = 0.011$). This differs from the positive long-run relationship, but is not

contradictory - short-run adjustments and long-run equilibrium effects can legitimately differ in direction.

Unlike the levels specifications, the changes-based model satisfies CLM assumptions (no autocorrelation or heteroskedasticity), though it has lower explanatory power ($R^2 = 0.336$).

4. Recommendations:

4.1 Design policies with a long-run horizon:

The positive and statistically significant Long-Run Propensity (LRP = 0.186) indicates that tax exemptions increase fertility over time. Therefore, such policies should be evaluated and implemented with a long-term perspective rather than expecting immediate effects.

4.2 Do not rely on short-run responses for policy evaluation:

The changes-based model shows a negative short-run effect (-0.0953), highlighting that immediate responses may differ from long-run outcomes. Policymakers should avoid judging effectiveness based on short-term fluctuations.

4.3 Ensure stability and predictability of tax policy:

The significance of lagged effects in the FDL model (joint F-test $p = 0.004$) implies that fertility decisions adjust gradually. Frequent policy changes may weaken their impact, making consistency crucial.

4.4 Complement fiscal incentives with broader socio-economic policies:

The strong negative effect of the education program (eduprog) suggests that structural factors significantly influence fertility. Tax policy alone is insufficient and

should be integrated with policies related to education, employment, and family support.

4.5 Use multiple evaluation frameworks in policy assessment:

Policy evaluations of fiscal incentives should draw on multiple model specifications, considering both long-run explanatory power and short-run statistical reliability, to avoid misleading conclusions from any single framework.

5. Conclusion and Limitations:

Conclusion:

This study examined the relationship between real tax exemptions and fertility in the United States from 1913 to 1984 using time-series econometric methods. The results indicate that tax exemptions have a statistically significant but modest short-run effect, while the long-run cumulative impact is positive and economically meaningful, as captured by the Long-Run Propensity (LRP).

The inclusion of lagged effects through the FDL model improves explanatory power and highlights the importance of dynamic adjustment in fertility behaviour. The changes-based model further confirms that short-run responses can differ from long-run outcomes. These findings underscore that fiscal incentives for fertility require sustained, long-horizon commitment to be effective.

Limitations:

Multicollinearity in the FDL model: High correlation among lagged tax variables limits the reliability of individual coefficient estimates.

Violation of CLM assumptions in levels models: The presence of autocorrelation and heteroskedasticity weakens inference in the static and FDL specifications.

Limited set of explanatory variables: Fertility is influenced by multiple factors (income, cultural shifts, labour market conditions), which are not fully captured in the model.

Data constraints: The use of aggregate annual data may mask heterogeneity across households and regions.

Model specification limitations: Potential structural breaks and nonlinear behavioural responses are not explicitly modelled.

6. References:

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